# Nathan Sauldubois

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## Summary


I am a postdoctoral researcher at **New York University (NYU)**, working on probability, optimal transport, martingale optimal transport, and model risk. I obtained my **PhD in Applied Mathematics from École Polytechnique** under the supervision of **Nizar Touzi** and defended my dissertation on **November 25, 2025**. My research combines theoretical probability, optimal control, optimal transport, and distributionally robust optimization.


## Skills

**Mathematical Interests:** Optimal transport, martingale optimal transport, probability, optimal control, mean-field games, robustness
**Numerical Interests:** Machine learning, deep learning for PDEs, entropic regularization, particle methods

**Programming:** Python, R, C++, LaTeX
**Scientific Computing:** Familiar with high-performance computing (HPC) and GPU computing
**Languages:** French (native), English (fluent)

## Publications

### Published and Accepted Papers

- Compoint, A., Sauldubois, N., and Touzi, N. (2026). *Sensitivity Analysis of Distributionally Robust BSDEs and RBSDEs*. *Mathematical Finance*.
- Sauldubois, N., and Touzi, N. (2026). *First-Order Martingale Model Risk and Semi-Static Hedging*. Accepted for publication in *SIAM Journal on Financial Mathematics*.

### Preprints

- Sauldubois, N. (2026). *Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach*. arXiv preprint arXiv:2601.20139.
- Sauldubois, N. (2026). *Higher-Order Sensitivity Analysis of Wasserstein and Adapted Wasserstein Distributionally Robust Optimization Problems*. arXiv preprint arXiv:2609.25345.
- Sauldubois, N., and Zhang, X. (2026). *Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport*. arXiv preprint arXiv:2605.30560.

### In Preparation

- Feng, Q., Sauldubois, N., and Zhang, X. (2026). *Initialization Stability and Convergence of Projected Langevin Dynamics*. Manuscript in preparation.

## Talks

- "A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem", Séminaire Doctorant du CMAP (2023)
- "Extended L-Q Mean-Field Game with Common Noise", Hammamet "Stochastic Control and Risk" (April 2023)
- "Robust Hedging and Distributionally Robust Sensitivity", PhD Talk, Séminaire Bachelier (March 2024)
- "Model Risk and Semi-Static Hedging", XII Bachelier World Congress, Rio (July 2024); and 7th Berlin Workshop on Mathematical Finance for Young Researchers, Berlin (September 2024)
- "Robust Static Hedging and Distributionally Robust Sensitivity", Hammamet Workshop, "Stochastic Control and Games for Risk and Regulation" (October 2024)
- "Adapted Wasserstein Gradient Flow and Projected Langevin Dynamics for Weak Optimal Transport", Hammamet "Stochastic Control and Risk" (April 2026)
- "Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach", XIII Bachelier World Congress, University of Bologna (July 1, 2026)
- "Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport", DMV Annual Meeting 2026, University of Konstanz (September 9, 2026); and 10th Eastern Conference on Mathematical Finance, Johns Hopkins University, Baltimore (October 2026)
- Invitation to the Third Young Mathematician Forum at NUS (January 11–12, 2027)

## Teaching

**NYU Tandon School of Engineering, M.S. in Financial Risk Engineering** **2026**

- **Option Pricing & Stochastic Calculus** — Graduate course (Spring and Fall 2026)
- **C++: Financial Computing Recitations** — Graduate recitations (Fall 2026)
- **Summer Bootcamp** (Summer 2026) and **Winter Math Bootcamp: From Brain Teasers to Black–Scholes** (Winter 2026)

**École Polytechnique** **2022–2025**

- **Stochastic Models in Finance** — Tutor and homework grader
- **Convex Optimization and Control** — Teaching assistant, third-year bachelor's course
- **Probability Refreshers** — Teaching assistant, X–HEC Master's in Data Science for Business

**Lycée du Parc** **2019–2020**

- **Preparatory Classes (CPGE)** — Oral exam preparation tutor

**Supervision**

- Co-supervision of **Arthur Compoint**'s research internship on *continuous-time model risk for stochastic optimization problems* (2025).
- Currently co-supervising **Tianshi Liu**, who started his PhD at NYU under the supervision of Xin Zhang, on *Numerical Scheme for Projected Langevin Dynamics* (2026).

## Administrative Responsibilities

- Communications and website maintenance, **Séminaire Bachelier** (CMAP–CREST)
- Co-organizer of the **Séminaire Doctorant du CMAP**, 2023–2024
- Co-organizer, with Xin Zhang, of the invited session "**Optimal Transport in Robust Finance**" at the 2026 INFORMS Conference on Financial Engineering and FinTech

## Education

**École Polytechnique** **2022–2025**
*PhD in Applied Mathematics, defended on November 25, 2025. Supervisor: Nizar Touzi.*
**ENS de Lyon** **2018–2021**
*L3 Fundamental Mathematics with a Physics Minor; M1 and M2 Advanced Mathematics*
**Special Jury Prize, SMF Junior Contest** **2018**

## Research Experience

**Postdoctoral Researcher, NYU** **2025–Present**

- Distributionally robust optimization in continuous time.
- Adapted Wasserstein gradient flows for entropic weak and martingale optimal transport.

**CMAP, École Polytechnique, PhD research** **2022–2025**

- Mean-field games with control-dependent interactions.
- Sensitivity analysis and distributional robustness in stochastic control.
- Martingale optimal transport and model risk.

**Natixis – Quantitative Research, internship supervised by Pierre Henry-Labordère** **2021–2022**

- Model uncertainty in fixed-income products.
- Super-replication strategies for Bermudan swaptions with non-constant notionals.
- Quantum optimal transport methods for correlation swaps.

**Earlier Research Internships**

- **CMAP** (2021, with Nizar Touzi and Hadrien De March): entropic optimal transport for healthcare policy design, with Quantev.
- **Oxford University** (2020, with Julien Berestycki): extremal particles of high-dimensional branching Brownian motion.
- **Université de Bordeaux** (2019, with David Lannes): shallow-water wave equations in bounded-depth domains.